ADMBD

Database Administration

Practical Assignment: Database Administration Project

Project requirements for Iteration 1

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1 Objective

The goal of this assignment is to apply and consolidate knowledge related to database administration, through the development of a medium-complexity project to be completed in teams. The assignment aims to enhance students' skills in designing, deploying, managing, and securing both **SQL** and **NoSQL** database systems in the context of a real-world application scenario.

The project is developed progressively in two iterations. **Iteration 1** focuses primarily on specification, analysis, architecture, and planning, while **Iteration 2** builds upon these decisions through implementation, testing, tuning, security configuration, backup and recovery, and final demonstration.

2 Delivery Deadlines and Team Info

- **Team size:** 3 to 5 students
- **Project type:** Team-based
- **Working model:** Work is to be carried out both during lab classes and outside scheduled hours.
- **Iteration 1 deadline:** November 8th (week 9)
- **Final submission (Iteration 2):** December 18th (week 14)

A portion of the lab classes will be used to support project development. All team members are expected to participate actively in the project and should be able to explain and justify the decisions and work carried out by the team.

3 Project Description

Each team must define and develop a project in a chosen application area, such as:

- Online Store or Marketplace
- Ride-hailing Platform (TVDE)
- Social Media System
- Learning Management System
- Healthcare or Hospital Management System
- Smart Home or IoT Platform

- Event or Training Management System
- Sports League or Tournament Manager

The selected domain must allow the use of both **SQL and NoSQL technologies**.

The project does not need to implement a complete production system. However, the proposed application must be sufficiently rich to support the database administration topics addressed throughout the course.

The project description should clearly identify the application domain, its motivation, the main users or actors, the main functional requirements, the relevant data to be managed, and the typical database operations expected in the application.

4 Project Repository and Documentation

Each team must create and maintain a **GitHub repository for the project**. The repository is mandatory and must be used throughout the complete development of the project, from **Iteration 1** until the final submission.

The repository must contain all relevant artefacts produced by the team, including, whenever applicable:

- Database schema and creation scripts;
- Configuration files and scripts;
- Data seeding scripts or notebooks;
- Queries, indexing, and tuning scripts;
- Backup and recovery scripts;
- Diagrams and other technical artefacts;
- Project documentation;
- Any source code developed in the context of the project.

The **source documentation must be stored in the repository in an editable, text-based format**, using either **LaTeX** or **Markdown**. Whenever a PDF report is required for submission, it should be generated from the documentation maintained in the repository.

The repository should be updated regularly by all students, throughout the project development. Its history should provide evidence of the evolution of the work and of the contributions made by the different team members.

The teaching staff must be granted access to the repository using their institutional **GitHub** accounts/email addresses. The teaching staff must have access to the repository throughout the development and assessment period of the project.

The repository **URL** must be included in the **Iteration 1** report and in the final report.

At the end of **Iteration 1**, the team must create a `git tag` named `iteration1`. This tag must identify the exact repository version corresponding to the **Iteration 1** submission and must be pushed to **GitHub** before the **Moodle** submission deadline.

5 Iteration 1 – Specification and Planning (due November 8th)

In this phase, the focus is on analysis, architecture, and planning. Each team must submit a report with the following artefacts:

1 Project Description and Requirements:

- Description and motivation of the chosen application domain;
- Identification of the main users or actors and relevant use cases;
- Identification of the main functional and data requirements;
- Initial identification of typical database operations and workloads.

2 Selection of Database Technologies:

- At least one relational (**SQL**) and one non-relational (**NoSQL**) **DBMS**;
- Justification of the choices based on the project context and requirements;
- Identification and brief discussion of relevant alternatives considered by the team.

3 Database Models Specification:

- Initial conceptual models or data diagrams (ER, document structure, graph model, etc., as applicable);
- Description of the main entities/data structures, attributes, relationships, identifiers, and relevant constraints;
- Clear identification of which data is managed by each selected database technology.

4 **Deployment and Architecture Analysis:**

- Initial architecture diagram identifying the main application and database components;
- Comparative research: on-premises *vs.* cloud deployment;
- Justification of the deployment approach chosen, even if the prototype will not use this approach;
- Identification of relevant distribution, replication, availability, or infrastructure requirements, when applicable.

5 **Backup Policy Specification:**

- Planned strategy for backing up both **SQL** and **NoSQL** data;
- Backup frequency, tools, formats, and retention considerations;
- System recovery outline and relevant recovery objectives/metrics.

6 **Scalability and Elasticity Plan:**

- Anticipated needs for scalability (read/write load, data growth, etc.);
- Conceptual plan for horizontal and/or vertical scaling in both **DBMSs**;
- Identification of the main technical considerations and trade-offs associated with the proposed approach.

7 **Project Repository:**

- **URL** of the project repository;
- Confirmation that the teaching staff have been granted access using their institutional contact details;
- Project documentation available in the repository in LaTeX or Markdown format;
- A Git tag named `iteration1`, created at the end of the iteration and pushed to **GitHub**, identifying the exact version submitted for **Iteration 1**.

The **Iteration 1** deliverables must be submitted via **Moodle** by the deadline and will be presented in class for grading and feedback. Only one submission per group is required.

The **Moodle** submission must contain the following **two files**:

- `ADMBD_groupXX_it1.zip` — the `ZIP` archive of the repository obtained directly from GitHub, corresponding to the repository version identified by the `iteration1` tag;

- `ADMBD_groupXX_it1.pdf` — the `PDF` report generated from the documentation sources stored in the repository.

In both filenames, `XX` must be replaced by the **group number assigned to the team**. The contents of the **ZIP** archive, downloaded from **GitHub** for the **iteration1** tag, and the **PDF** report must correspond to the repository state identified by the `iteration1` tag.

The submission must include a **Group Self-Evaluation**. For each team member, identified by student number, the group must assign one of the following ratings to each member (only one per person): "Did not work"; "Far below average"; "Slightly below average"; "Average"; "Slightly above average"; "Far above average"; "Did all the work".

The ratings represent each member's contribution relative to the other members of the team, not relative to students in other groups or to the expected performance of an average student. Consequently, if all team members contributed equally, all should be rated "Average". If one or more members are rated above average, at least one other member must be rated below average.

This self-assessment helps the teaching staff understand the internal team dynamics and may be used, together with the repository history and other evidence of participation, to adjust individual grades if necessary.

Example:

Student Number	Did not work	Far below avg	Slightly below avg	Average	Slightly above avg	Far above avg	Did all the work	Tasks the student was responsible for
20231234				■				DB Models and Backup
20231235					■			DB Technologies and DB Models
20231236		■						Backups and Scalability
20231237						■		Deployment approach and Scalability

6 Project Checkpoints and Evaluation

- **Checkpoint 1 (end of week 3):** Team and idea registration
- **Checkpoint 2 (Week 9):** Iteration 1 submission
- **Checkpoint 3 (Week 9–10):** Iteration 1 presentation in PL classes
- **Checkpoint 4 (Week 14):** Final submission
- **Checkpoint 5 (Week 15):** Demonstration and evaluation

7 Evaluation

Iteration 1 represents **30% of the project evaluation**, while **Iteration 2** represents the remaining **70%**.

The detailed assessment criteria, component weights, and grading descriptors for both iterations will be **published on Moodle**. They are intentionally not reproduced in this document.

Students should therefore consult **Moodle** for the current evaluation criteria applicable to each iteration.

8 Use of Generative AI Tools

The use of **Generative Artificial Intelligence (GenAI) tools**, such as **ChatGPT**, **GitHub Copilot**, **Gemini**, **Claude**, or similar tools, is permitted as a supporting resource during the development of the project.

These tools may be used, for example, to support brainstorming, improve documentation, assist with code or query development, explain technical concepts, or help identify possible solutions. However, their use does not replace the team's responsibility for the design, implementation, documentation, and validation of the submitted work.

All content produced or suggested by **GenAI** tools must be **critically reviewed, validated, and adapted by the team**. Students are fully responsible for the correctness, quality, security, and technical adequacy of all submitted artefacts, regardless of whether **GenAI** tools were used in their production.

In particular:

- Students must understand and be able to explain all submitted code, queries, configurations, diagrams, architectural decisions, and documentation;
- **GenAI**-generated output must not be included in the project without appropriate verification and adaptation to the specific project context;
- **GenAI** tools must not be used as a substitute for the technical analysis, experimentation, benchmarking, or justification required by the assignment;
- Information obtained through **GenAI** tools must not be treated as an authoritative source. Whenever appropriate, technical claims should be verified using official documentation, scientific literature, or other reliable sources;
- Confidential information, credentials, passwords, access tokens, or other sensitive project information must not be submitted to external **GenAI** services.

Declaration of AI Usage

The project documentation must contain a short section entitled **"Use of Generative AI"** describing how **GenAI** tools were used during the corresponding iteration.

For each relevant use, the team must identify:

- The tools used;
- The purpose for which they were used;
- The project artefact(s) or task(s) in which they were used;
- How the generated output was reviewed, validated, or modified by the team.

If no **GenAI** tools were used, this must also be explicitly stated in that section.

The team remains responsible for all submitted material and all team members must be able to explain and defend the work during the project presentations and evaluation sessions. Inability to explain or justify submitted material may be taken into account in the assessment, independently of whether that material was produced with the assistance of **GenAI** tools.

Final Remarks

The success of the project depends on both the technical quality of the deliverables and the effectiveness of team collaboration. Each member's contribution must be clearly identifiable and properly documented throughout the report, repository, and project artefacts. Significant discrepancies in participation or commitment may result in differentiated individual grades.

The report should focus on the approach actually adopted by the team and on the justification and evidence of the work performed. It should avoid unnecessary theoretical explanations of concepts such as scalability, elasticity, database models, or backup types. The emphasis should be on how those concepts apply to the proposed system and, in **Iteration 2**, how they were implemented and validated.

Note: The grade assigned to **Iteration 1** represents 30% of the final project grade but is considered provisional until completion of Iteration 2. If the project demonstrates significant improvements or meaningful extensions of the **Iteration 1** components during **Iteration 2**, the corresponding part of the **Iteration 1** evaluation may be slightly revised upward to reflect this progress.